Database Fundamentals Project:

1. Identify the relevant entities of the hospital dataset with their respective attributes, entity types (strong or weak) and primary keys. (8 Marks)

The relevant entities in this database are medical department, treatments, staff, patients, appointments and room where all of them are strong entities, and their primary keys are:

Department: department_ID

Treatment: treatment_ID

Staff: staff_ID

Patient: patient_ID

Appointment: appointment_ID

Room: room_ID

And their attributes are:

Department: department_ID, name, location, contact_number

Treatments: treatment_ID, name, description, cost, department_ID

Staff: staff_ID, name, role, contact_number, email, department_ID, hire_date

Patients: patient_ID, name, date_of_birth, contact_number, address, medical_history, room_ID.

Appointment: appointment_ID, patient_ID, Staff_ID, treatment_ID, date, time, notes

Room: room_ID, type, capacity, availability.

2. Clearly explain why each relation (entity) needs to be included (8 Marks)

Each relation must be included in the database because:

Departments: a hospital is composed of plenty of different departments each one with its own resource, treatments and staff. This entity allows the database to store all departments.

Treatments: this entity registers all treatments offered by the hospital, it allows to relate departments and treatments at the same time it makes easier monitoring.

Staff: a hospital needs staff to work, this entity allows to know who does the treatment and other data for it such as costs, descriptions and the department where its taking place.

Patient: patients are the core of the hospitals database, they are needed in order to keep working.

Appointment: this entity registers when a patient goes to the hospital, the doctor that attends the patient, why is the patient at the hospital and it helps organizing the patient's appointment and keeps a organized structure.

Room: its purpose is to help with the organization with the admitted patients, it controls the disponibility of rooms and it makes the rooms assignment easier.

3. Identify the relationships, cardinalities and participation constraints with supporting business rules and assumptions. (8 Marks)

The relationships are the following ones:

- Departments – staff relationship: it is a 1-N relationship because each person must be associated with a single department but a department can have more than one staff on it.

  Each department has at least 1 staff and each staff must be part of one department.

  Relation has total participation because all staff has to be assigned to a department.

- Departments – treatment: this is a 1-N relation. Each department offers at least 1 treatment and each treatment is part of one department. All treatments are associated with at least one department and one department can offer more than 1 treatments. This is partial participation because a medical department can exist without an assigned treatment.

- Staff – appointment: this is a 1-N relation. Each appointment must be attended by a health professional, and a doctor can attend different appointments at different time. it is partial participation because a staff can exist without having appointments.

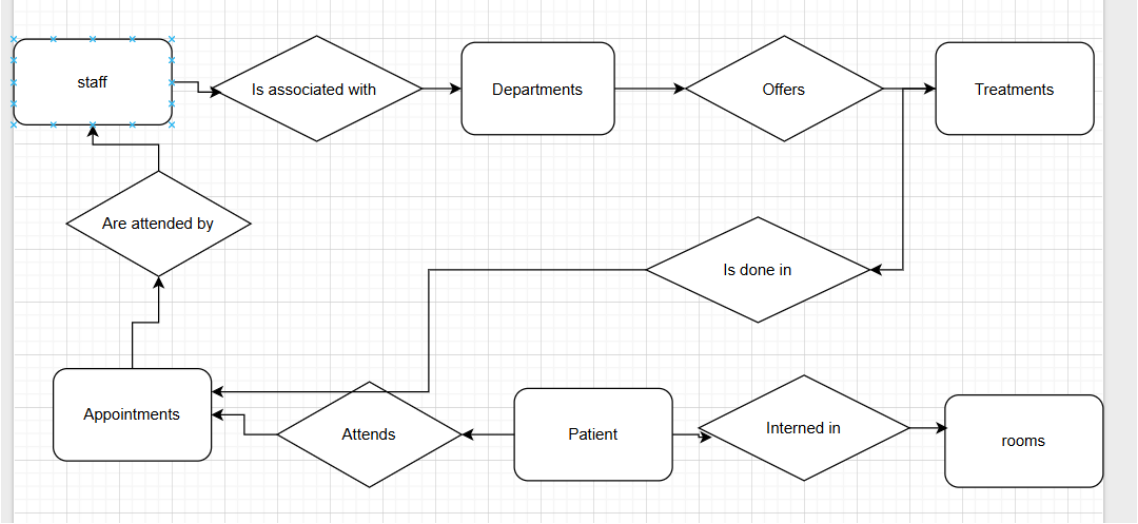
- Patient – appointment: this is a 1-N relation. Each appointment must be related to one patient but a patient can have different appointments in different dates. Not all appointments have dates and an appointment always is for a patient. This is a partial participation because a patient can exist without having an appointment.

- Treatment – appointment: this is a 1-N relation. All appointments must have a treatment on it but a single treatment can be used in different appointments. A treatment can still not be in any appointment but an appointment always corresponds to a specific treatment. This is a partial participation because a treatment can exist without an appointment but an appointment can not exist without the treatment.

- Patient – room: this is a 1-N relation. Rooms are for interned patients, a patient can only use one room at a time, and one room can take more than one patient at a time. the room can be empty and not every patient needs a room. This is a partial participation because a room can be empty but it is still there while a patient does not always need a room.

4. Draw conceptual level ER diagram for the system depicting the entities, relationships, cardinalities, participation using your preferred ERD notation. (4 Marks)

This is the ER diagram:



Here we can see the relationship between all entities and how they are related:

One staff is associated with at least one department and a department is associated with staff.

One department Offers one or more treatments while a single treatment can only be Offered in one department.

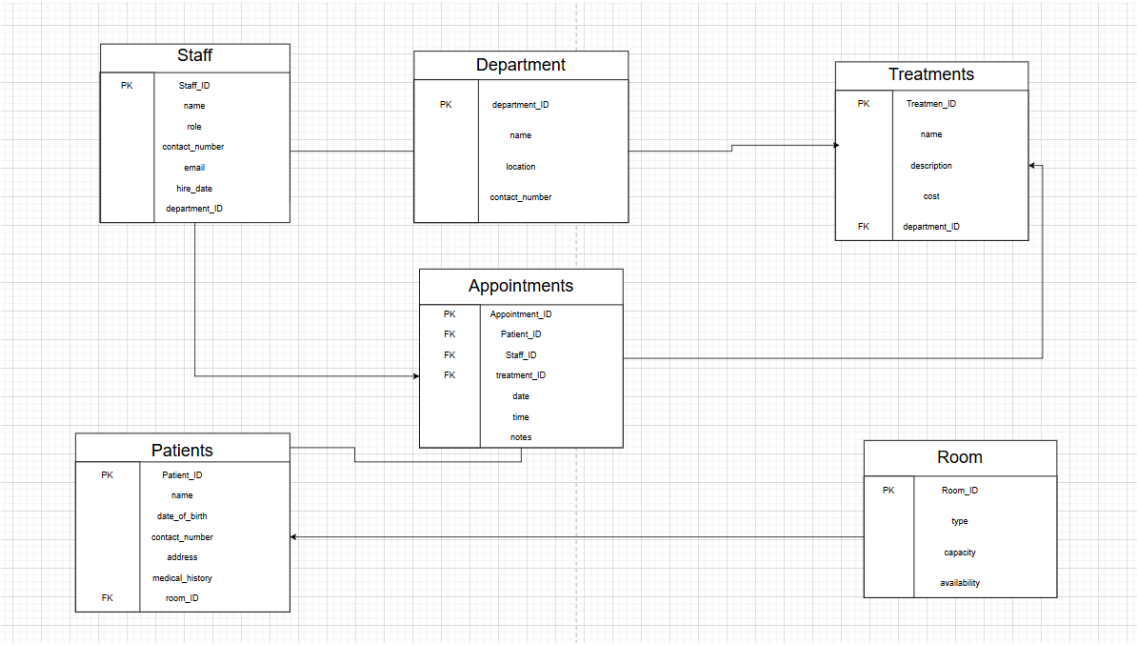
All appointments are attended by at least one staff member but a single staff member can attend different appointments during the day.

All appointments are attended by a patient while a patient can assist to different appointments scheduled in different hours.

A patient can only be using one room at a time but a room can be used by more than one person.

All appointments must have a treatment in it but the same treatment can be used in different appointments.

5. Convert the entities of conceptual design into logical dataset and draw logical ERD. Use a logical layout of entities which is easily understood by any reader. (4 Marks)



Here as we can see there are all the entities In the database whith their Primary key, Foreign key and also their attributes.

Staff: Staff_ID (PK), department_ID(FK)

Departments: department_ID (PK)

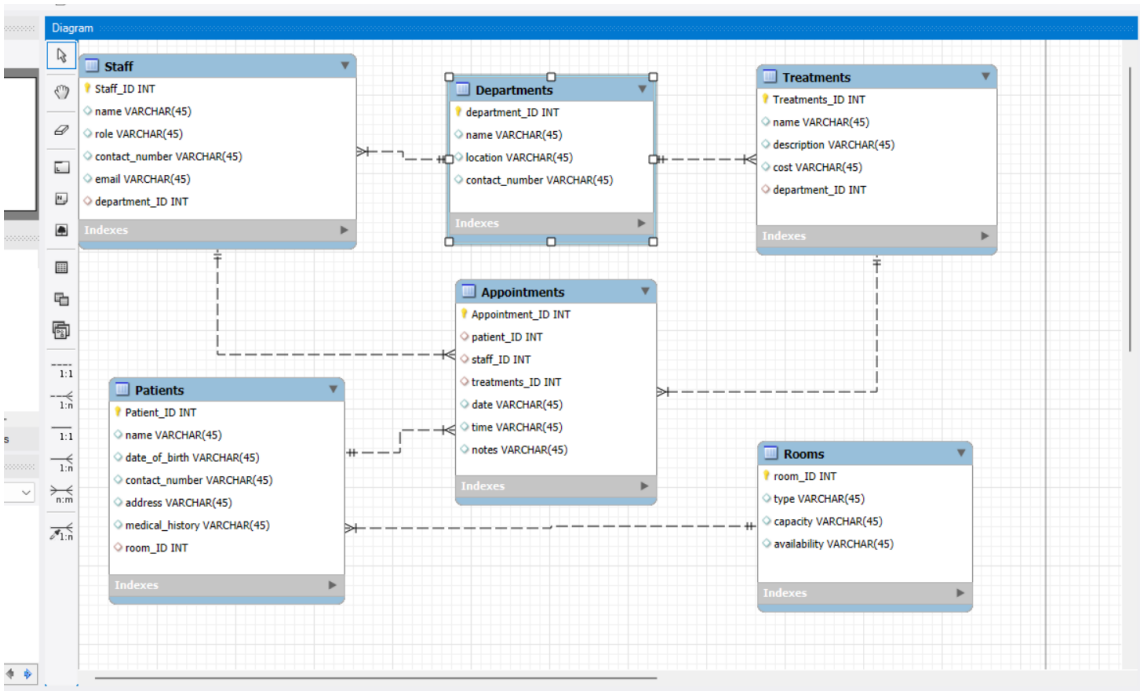
Treatments: treatment_ID (PK), department_ID(FK)

Appointments: Appointments_ID(PK), Patient_ID(FK), Staff(FK), treatment_ID(FK)

Patients: patients_ID(PK), room_ID(FK)

Room: room_ID(PK).

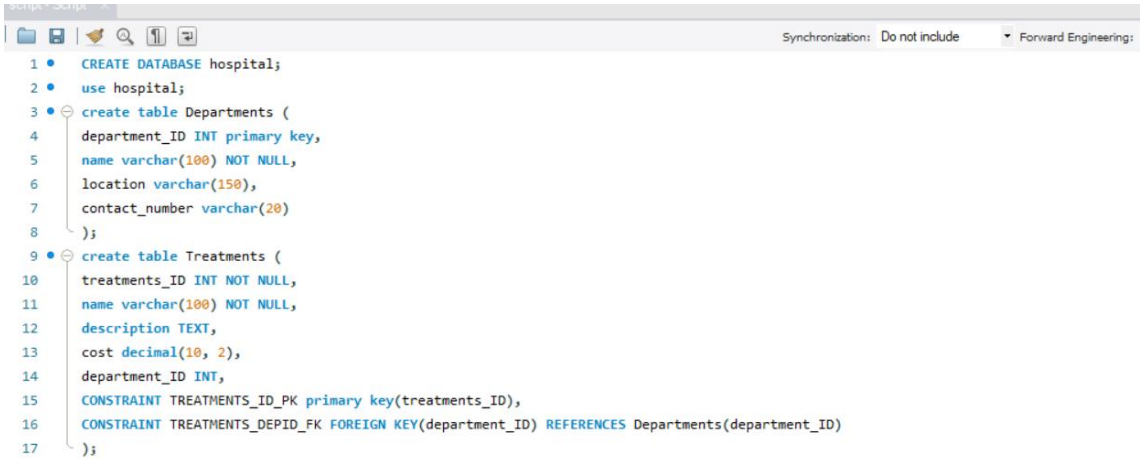
6. Convert the entities of the logical design into physical model and draw physical level ERD. (4 Marks)



7. Convert the physical level ERD into relational model. Ensure that tables are in 3rd Normal Form. (4 Marks)

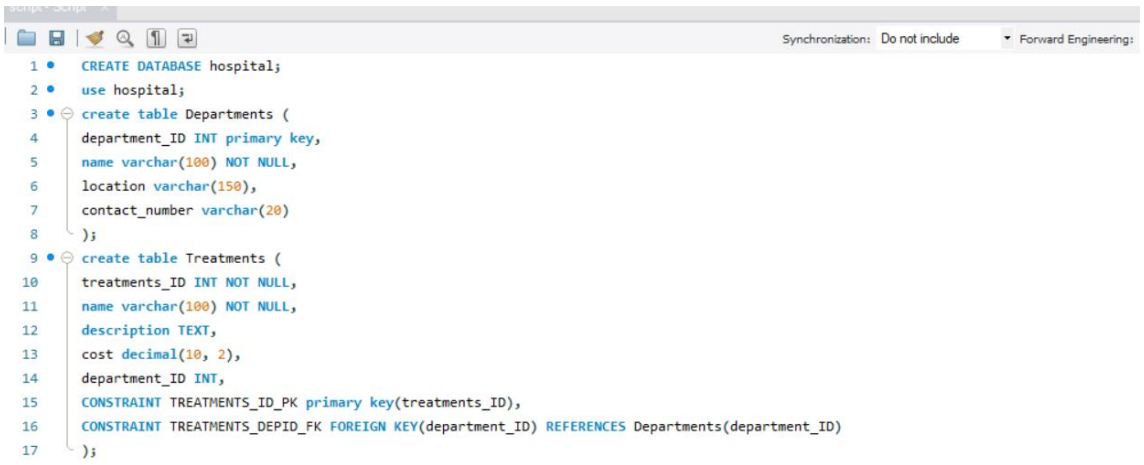
Task 2

1. Create the database and all tables specified in the dataset (relational model) to meet the specifications identified in task one. Clearly document all DDL statements used in this process. (10 Marks)



Synchronization: Do not include Forward Engineering:

```
1  CREATE DATABASE hospital;
2  use hospital;
3  create table Departments (
4  department_ID INT primary key,
5  name varchar(100) NOT NULL,
6  location varchar(150),
7  contact_number varchar(20)
8  );
9  create table Treatments (
10 treatments_ID INT NOT NULL,
11 name varchar(100) NOT NULL,
12 description TEXT,
13 cost decimal(10, 2),
14 department_ID INT,
15 CONSTRAINT TREATMENTS_ID_PK primary key(treatments_ID),
16 CONSTRAINT TREATMENTS_DEPID_FK FOREIGN KEY(department_ID) REFERENCES Departments(department_ID)
17 );
```



Synchronization: Do not include Forward Engineering:

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1  CREATE DATABASE hospital;
2  use hospital;
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15 CONSTRAINT TREATMENTS_ID_PK primary key(treatments_ID),
16 CONSTRAINT TREATMENTS_DEPID_FK FOREIGN KEY(department_ID) REFERENCES Departments(department_ID)
17 );
```

```sql
create table Patients (
patient_ID INT NOT NULL,
name varchar(100) NOT NULL,
date_of_birth DATE,
contact_number varchar(20),
address varchar(200),
medical_history TEXT,
room_ID INT NOT NULL,
CONSTRAINT PATIENTS_ID_PK primary key(patient_ID),
CONSTRAINT PATIENTS_ROOMID_FK foreign key(room_ID) REFERENCES Rooms(room_ID)
);
create table Appointments (
appointment_ID INT NOT NULL,
patient_ID INT NOT NULL,
Staff_ID INT NOT NULL,
treatments_ID INT NOT NULL,
date DATE,
time TIME,
notes text,
CONSTRAINT APPOINTMENTS_ID_PK primary key(appointment_ID),
CONSTRAINT APPOINTMENTS_PATID_FK FOREIGN KEY(patient_ID) REFERENCES Patients(patient_ID),
CONSTRAINT APPOINTMENTS_STAFFID_FK foreign key(Staff_ID) REFERENCES Staff(Staff_ID),
CONSTRAINT APPOINTMENTS_TREATID_FK foreign key(treatments_ID) references Treatments(treatments_ID)
);
```

2. Populate all database tables with the desired data to a minimum of 10 tuples per relation. Clearly document all DML statements used in this process (10 Marks)

```sql
use hospital;
INSERT INTO Departments VALUES
(1, 'Cardiology', 'Building A', '555-1001'),
(2, 'Neurology', 'Building B', '555-1002'),
(3, 'Pediatrics', 'Building C', '555-1003'),
(4, 'Emergency', 'Building A', '555-1004'),
(5, 'oncology', 'Building D', '555-1005'),
(6, 'Dermatology', 'Building E', '555-1006'),
(7, 'Orthopedics', 'Building F', '555-1007'),
(8, 'Gastroenterology', 'Building B', '555-1008'),
(9, 'urology', 'Building C', '555-1009'),
(10, 'Radiology', 'Building D', '555-1010'),
(11, 'Orthopedic Surgery', 'Building F', '555-1011');
```

```sql
INSERT INTO Treatments VALUES
(1, 'ECG', 'Heart activity test', 120, 1),
(2, 'MRI Scan', 'Magnetic resonance imaging', 900, 10),
(3, 'Skin Biopsy', 'Removal of skin sample', 150, 6),
(4, 'Chemotherapy', 'Cancer treatment', 750, 5),
(5, 'X-Ray', 'Bone imaging', 80, 10),
(6, 'Physiotherapy', 'Muscle rehabilitation', 60, 7),
(7, 'Gastrocopy', 'Stomach examination', 300, 8),
(8, 'Ultrasound', 'internal organ scan', 150, 10),
(9, 'Bloodtest', 'Routine hematology test', 40, 4),
(10, 'CT scan', 'Computed tomography scan', 500, 10),
(11, 'Heart surgery', 'Heart surgery', 600, 1);
INSERT INTO Staff VALUES
(1, 'Dr. Alice White', 'Doctor', '555-2001', 'alice.white@hospital.com', 1, '2025-01-10'),
(2, 'Dr. Ben Torres', 'Neurologist', '555-2002', 'ben.torres@hospital.com', 2, '1992-05-14'),
(3, 'Nurse Carla Diaz', 'Nurse', '555-2003', 'carla.diaz@hospital.com', 4, '1997-11-23'),
(4, 'Dr. Daniel Kim', 'Oncologist', '555-2004', 'daniel.kim@hospital.com', 5, '2001-03-08'),
(5, 'Nurse Eva Smith', 'Nurse', '555-2005', 'eva.smith@hospital.com', 3, '2005-07-19'),
(6, 'Dr. Frank Lee', 'Dermatologist', '555-2006', 'frank.lee@hospital.com', 6, '1994-12-30'),
(7, 'Tech Greg Hall', 'Radiology Tech', '555-2007', 'greg.hall@hospital.com', 10, '2008-01-17');
```

Output


```sql
INSERT INTO Treatments VALUES
(1, 'ECG', 'Heart activity test', 120, 1),
(2, 'MRI Scan', 'Magnetic resonance imaging', 900, 10),
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(5, 'X-Ray', 'Bone imaging', 80, 10),
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(7, 'Gastrocopy', 'Stomach examination', 300, 8),
(8, 'Ultrasound', 'internal organ scan', 150, 10),
(9, 'Bloodtest', 'Routine hematology test', 40, 4),
(10, 'CT scan', 'Computed tomography scan', 500, 10),
(11, 'Heart surgery', 'Heart surgery', 600, 1);
INSERT INTO Staff VALUES
(1, 'Dr. Alice White', 'Doctor', '555-2001', 'alice.white@hospital.com', 1, '2025-01-10'),
(2, 'Dr. Ben Torres', 'Neurologist', '555-2002', 'ben.torres@hospital.com', 2, '1992-05-14'),
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(7, 'Tech Greg Hall', 'Radiology Tech', '555-2007', 'greg.hall@hospital.com', 10, '2008-01-17');
```

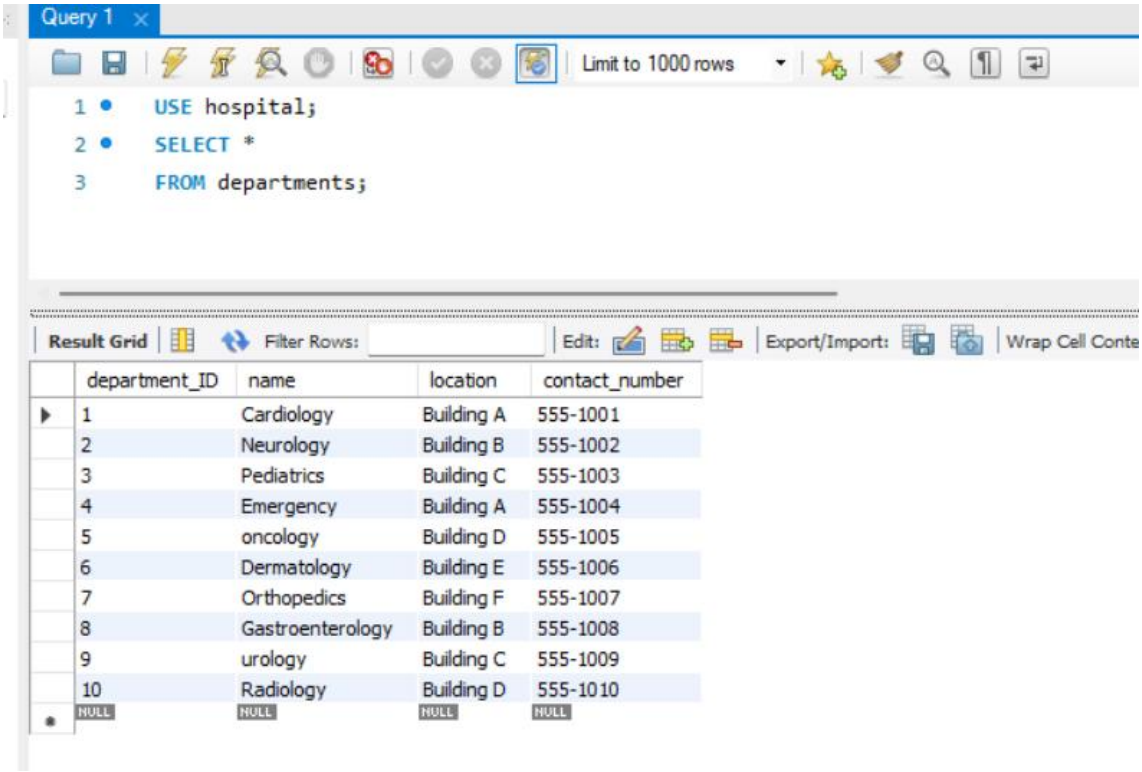
Output

```
INSERT INTO Rooms VALUES
(1, 'Single', 1, 1),
(2, 'Double', 2, 0),
(3, 'ICU', 1, 1),
(4, 'Single', 1, 0),
(5, 'Ward', 4, 1),
(6, 'Double', 2, 1),
(7, 'ICU', 1, 0),
(8, 'Ward', 6, 1),
(9, 'Single', 1, 1),
(10, 'Double', 2, 0);
```

```
INSERT INTO Patients VALUES
(1, 'John Carter', '1985-04-12', '555-3001', 'Street 12A', 'Diabetes', 1),
(2, 'Mary Lopez', '1992-07-20', '555-3002', 'Avenue 45B', 'Hypertension', 2),
(3, 'Peter Wong', '1978-01-05', '555-3003', 'Block C14', 'Asthma', 3),
(4, 'Linda Green', '2001-09-19', '555-3004', 'Lane 22', 'Healthy', 4),
(5, 'Alex Rivera', '1965-11-13', '555-3005', 'Street 89', 'Cancer', 10),
(6, 'Chloe Smith', '1999-12-30', '555-3006', 'Hill Road 9', 'Dermatitis', 6),
(7, 'Brian Fox', '1983-06-21', '555-3007', 'South Ave 3', 'Broken Leg', 5),
(8, 'Sofia Torres', '1975-02-15', '555-3008', 'Forest Way', 'Gastritis', 8),
(9, 'Marco Ruiz', '2004-10-10', '555-3009', 'Tower 4', 'Healthy', 7),
(10, 'Emily Brown', '1990-08-03', '555-3010', 'Lake St', 'Hypertension', 9),
(5612, 'John Bob', '1996-02-08', '123-456-7890', 'Calle Canal', 'Diabetes', 1),
(3456, 'Elena Rosa', '2000-01-10', '622-624-896', 'Calle La Haza', 'Migrains', 3);
INSERT INTO Appointments VALUES
(1, 1, 1, 11, '2025-01-10', '09:00', 'Patient reported mild headache; recommended rest and hydration'),
(2, 2, 2, 2, '2025-01-12', '10:30', 'Follow-up after surgery; stitches healing properly'),
(3, 3, 3, 9, '2025-01-13', '11:00', 'Prescribed antibiotics for 7 days; monitor for side effects'),
(4, 4, 1, 5, '2025-01-15', '12:00', 'Patient requested appointment reschedule due to work conflict'),
(5, 5, 4, 1, '2025-01-18', '14:30', 'Blood pressure slightly elevated; advised diet and exercise' ),
(6, 6, 6, 3, '2025-01-20', '15:00', 'Patient allergic to penicillin; alternative medication prescribed'),
(7, 7, 8, 6, '2025-01-22', '16:00', 'Routine check-up; all vitals within normal range'),
(8, 8, 9, 7, '2025-01-25', '09:15', 'Discussed treatment plan and consent forms signed'),
(9, 9, 7, 8, '2025-01-27', '13:45', 'Patient complained of recurring back pain; referred to physiotherapy'),
(10, 10, 2, 10, '2025-01-30', '10:00', 'First consultation; medical history reviewed, lab tests ordered')
```

TASK 3.

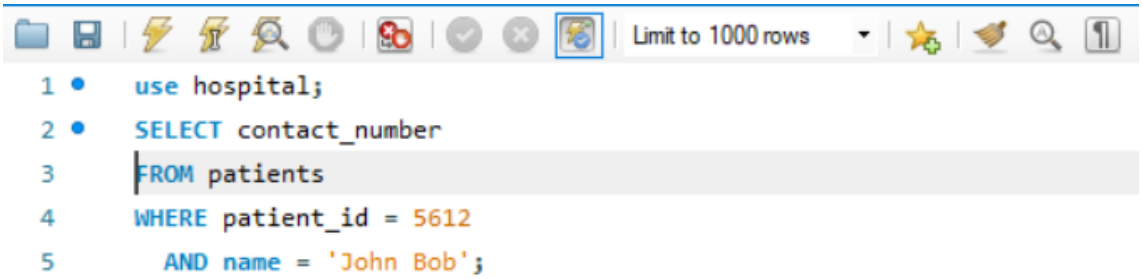
1. Show details of all medical departments of hospital. (2 Marks)

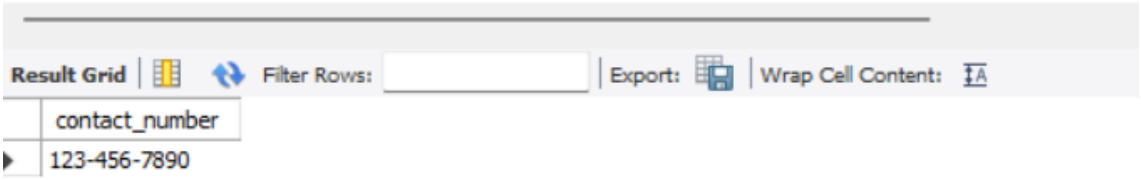


department_ID	name	location	contact_number
1	Cardiology	Building A	555-1001
2	Neurology	Building B	555-1002
3	Pediatrics	Building C	555-1003
4	Emergency	Building A	555-1004
5	oncology	Building D	555-1005
6	Dermatology	Building E	555-1006
7	Orthopedics	Building F	555-1007
8	Gastroenterology	Building B	555-1008
9	urology	Building C	555-1009
10	Radiology	Building D	555-1010
NULL	NULL	NULL	NULL

Here I use Select * to access all information about department table.

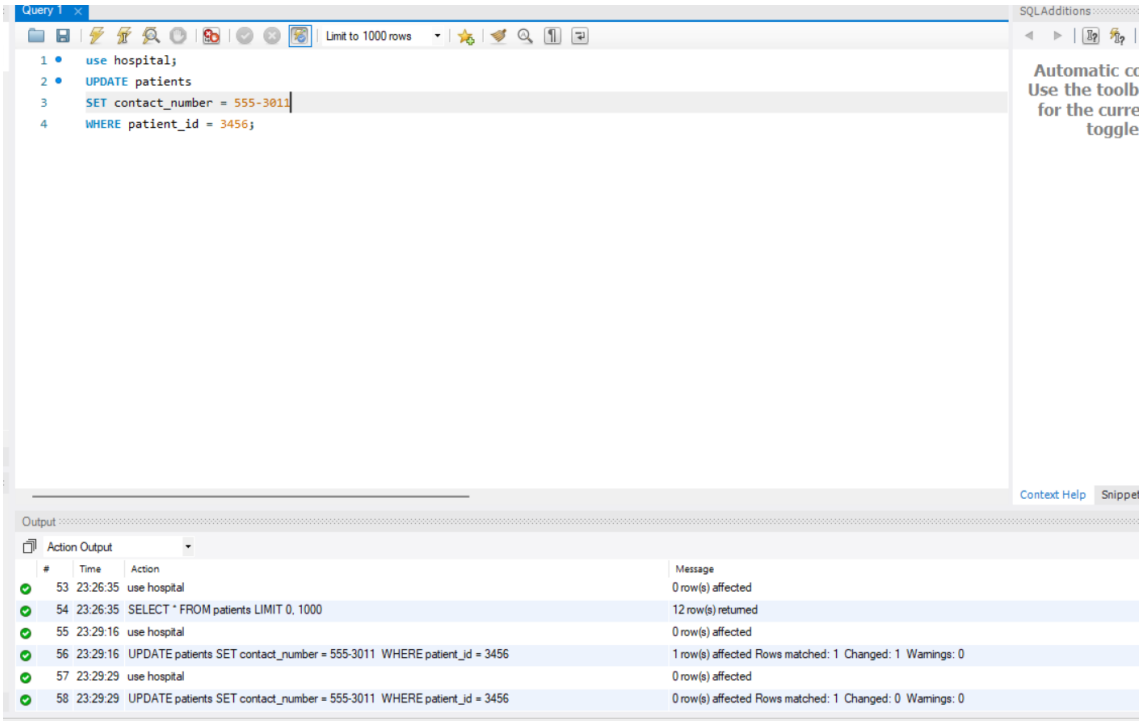
2. Find the contact details of patient “John Bob” whose patient ID is 5612.(2 Marks)





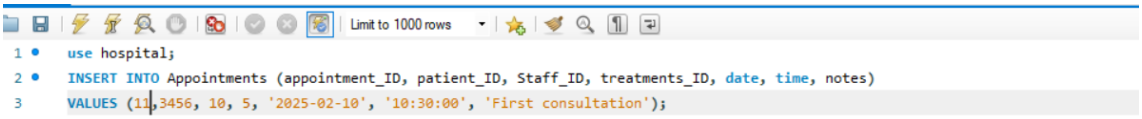
In this exercise, I used the statement WHERE in order to search for patient_ID 5612 and the person's name to get his contact information.

3. Update the patient ID number 3456 contact details. (2 Marks)



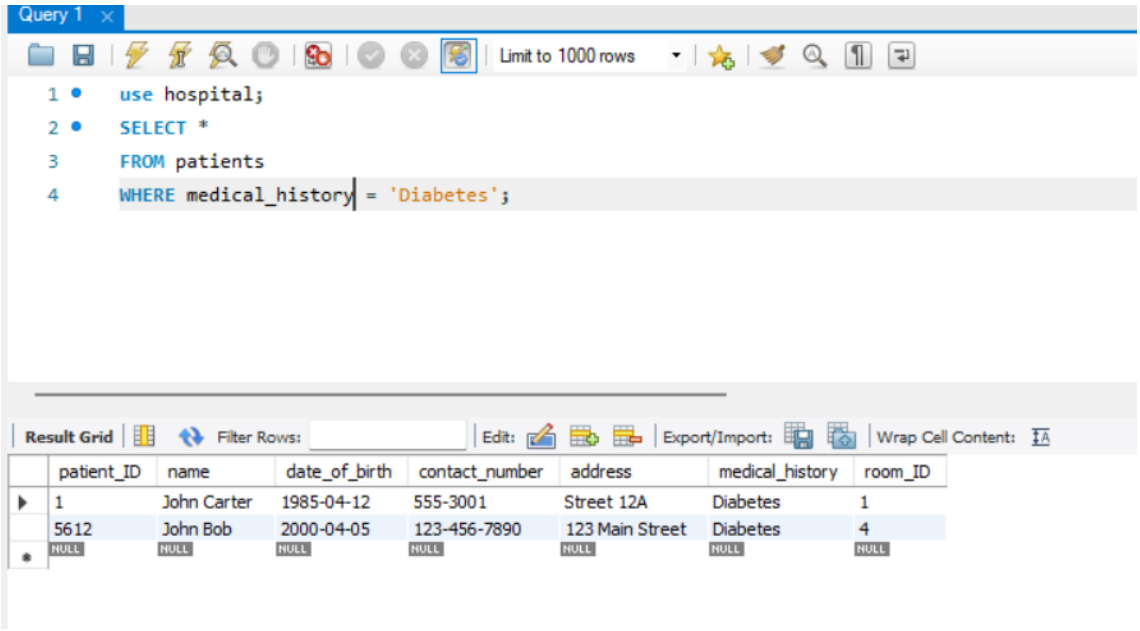
In this example I use UPDATE statement un order to change patient_ID 3456 contact number to another phone number.

4. Schedule a new medical appointment. (2 Marks)



In this exercise, I use INSERT statement in order to be able to access appointments table and add a new row using the tables column's names.

5. Find details of all patients who have Diabetes. (2 Marks)



In this exercise, I select all the columns (SELECT *) from table 'Patients' and look for people that have Diabetes using WHERE statement and searching in medical_history column.

6. Display the first name and surname of all patients who are under the care of Dr. Kavin Smith. (2 Marks)

7. Change all patients in Orthopedics department to Orthopedic Surgery Department. (2 Marks)

8. Display the first name, surname, and hire date of all doctors sorted in descending order by hire date. (2 Marks)

```
use hospital;
SELECT name, hire_date
FROM Staff
WHERE role = 'Doctor'
ORDER BY hire_date DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name	hire_date
Dr. Alice White	2025-01-10

In this exercise, I use select to have as an output the name (which includes the surname) and the date that the person with Doctor role was hired, it is selected from the Staff table and I use order by to make the order descendant.

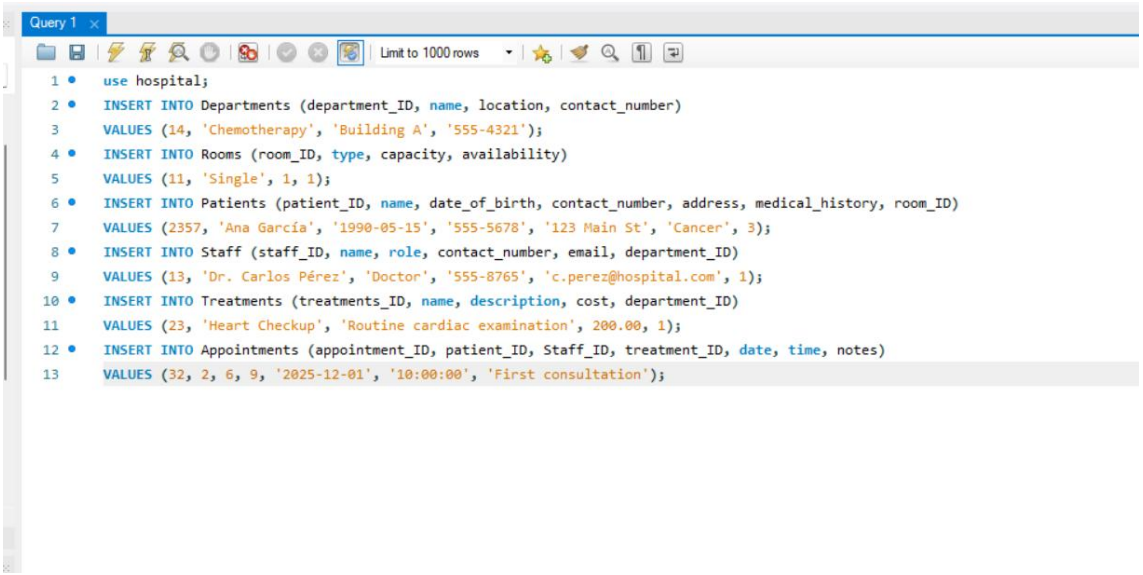
9. Count how many patients are undergoing 'heart Surgery' treatment. (2 Marks)

```
USE hospital;
SELECT COUNT(DISTINCT patient_ID) AS total_patients
FROM APPOINTMENTS
WHERE treatments_ID = 11;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_patients
1

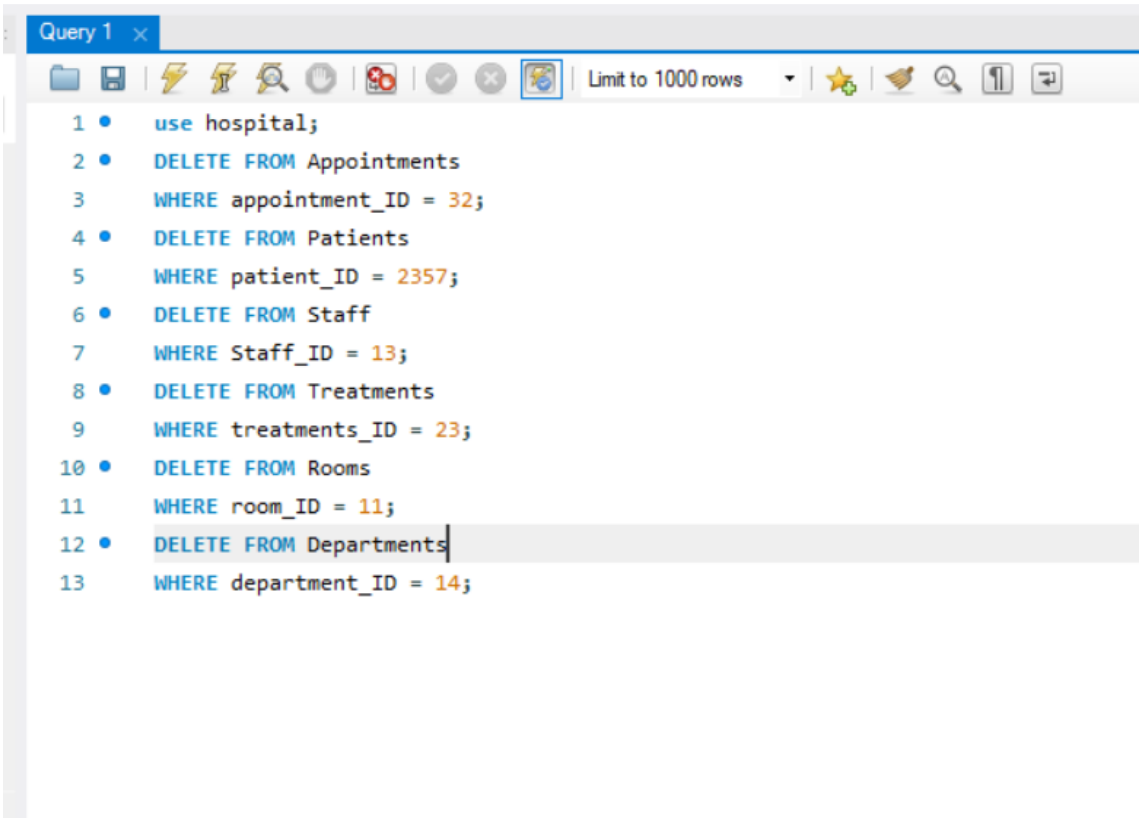
10. Add one new record to each table. (2 Marks)



```
use hospital;
INSERT INTO Departments (department_ID, name, location, contact_number)
VALUES (14, 'Chemotherapy', 'Building A', '555-4321');
INSERT INTO Rooms (room_ID, type, capacity, availability)
VALUES (11, 'Single', 1, 1);
INSERT INTO Patients (patient_ID, name, date_of_birth, contact_number, address, medical_history, room_ID)
VALUES (2357, 'Ana García', '1990-05-15', '555-5678', '123 Main St', 'Cancer', 3);
INSERT INTO Staff (staff_ID, name, role, contact_number, email, department_ID)
VALUES (13, 'Dr. Carlos Pérez', 'Doctor', '555-8765', 'c.perez@hospital.com', 1);
INSERT INTO Treatments (treatments_ID, name, description, cost, department_ID)
VALUES (23, 'Heart Checkup', 'Routine cardiac examination', 200.00, 1);
INSERT INTO Appointments (appointment_ID, patient_ID, Staff_ID, treatment_ID, date, time, notes)
VALUES (32, 2, 6, 9, '2025-12-01', '10:00:00', 'First consultation');
```

In this example I use INSERT statement in order to insert information into the tables, more specifically one row per table where I use all its columns in order to put information into it using VALUES statement.

11. Delete one record from all tables. (2 Marks)

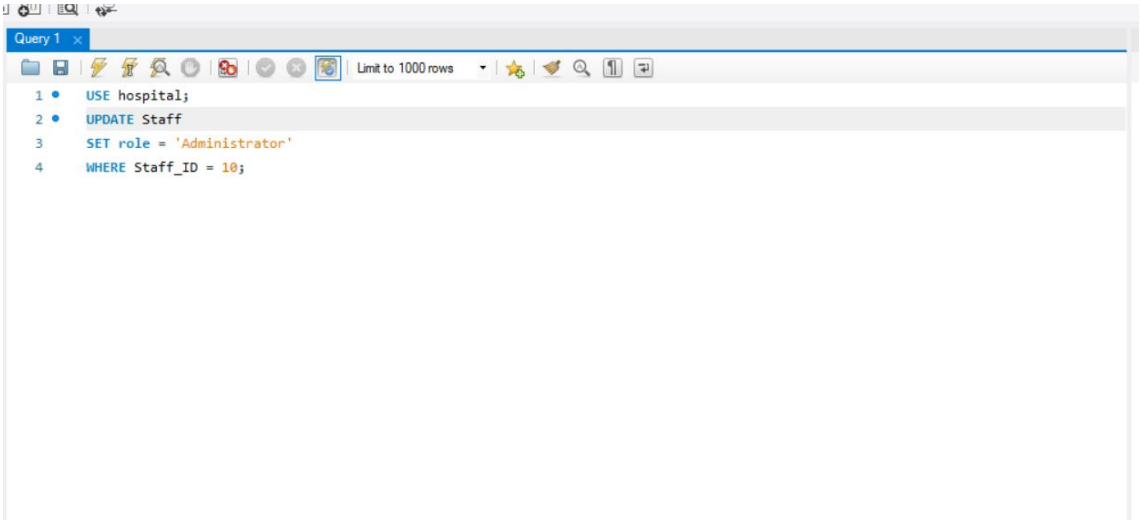


```
use hospital;
DELETE FROM Appointments
WHERE appointment_ID = 32;
DELETE FROM Patients
WHERE patient_ID = 2357;
DELETE FROM Staff
WHERE Staff_ID = 13;
DELETE FROM Treatments
WHERE treatments_ID = 23;
DELETE FROM Rooms
WHERE room_ID = 11;
DELETE FROM Departments
WHERE department_ID = 14;
```

In this exercise I used DELETE to eliminate the information that I added in the previous exercise using the ID as reference.

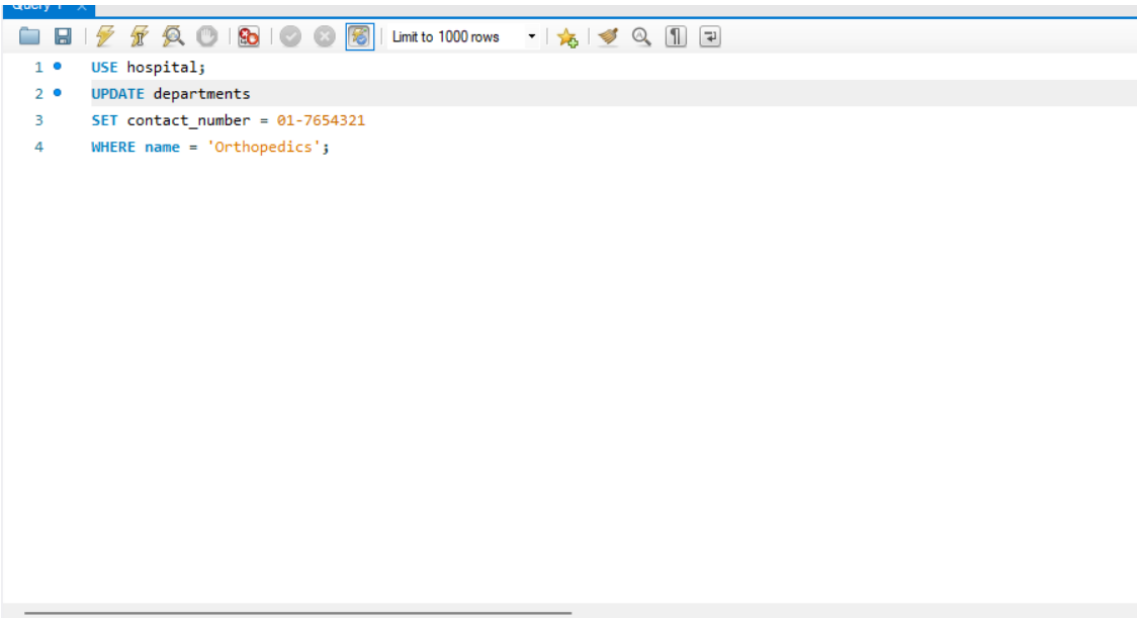
12. Find the total number of days off for all staff, order this by least days off. (2 Marks)

13. Change the roll of all staff who have a job title Office worker to Administrator. (2 Marks)

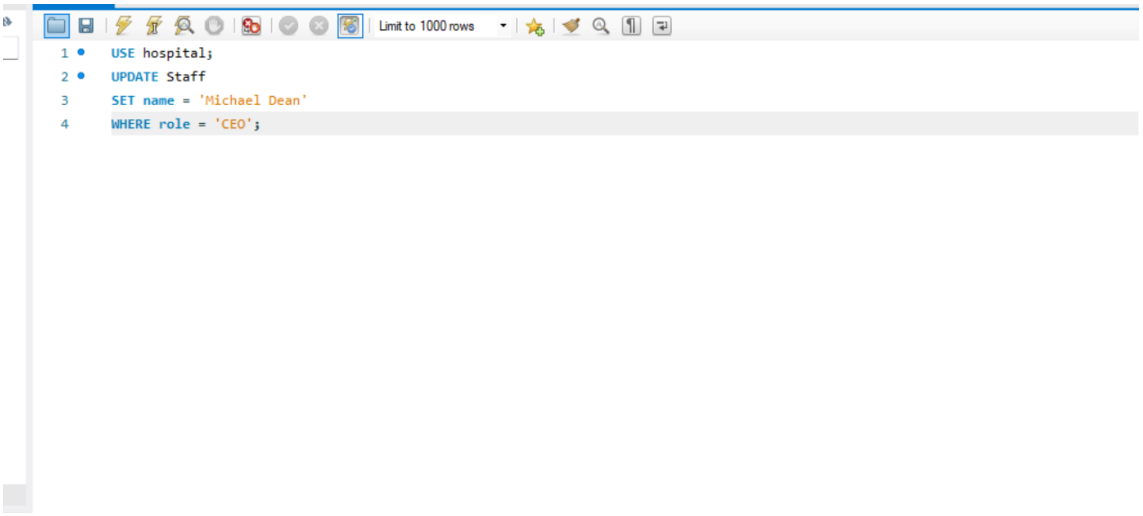


In this exercise, I use UPDATE with Staff_ID in order to change the staff role.

14. Set the Orthopedics department phone number to be 01-7654321. (2 Marks)



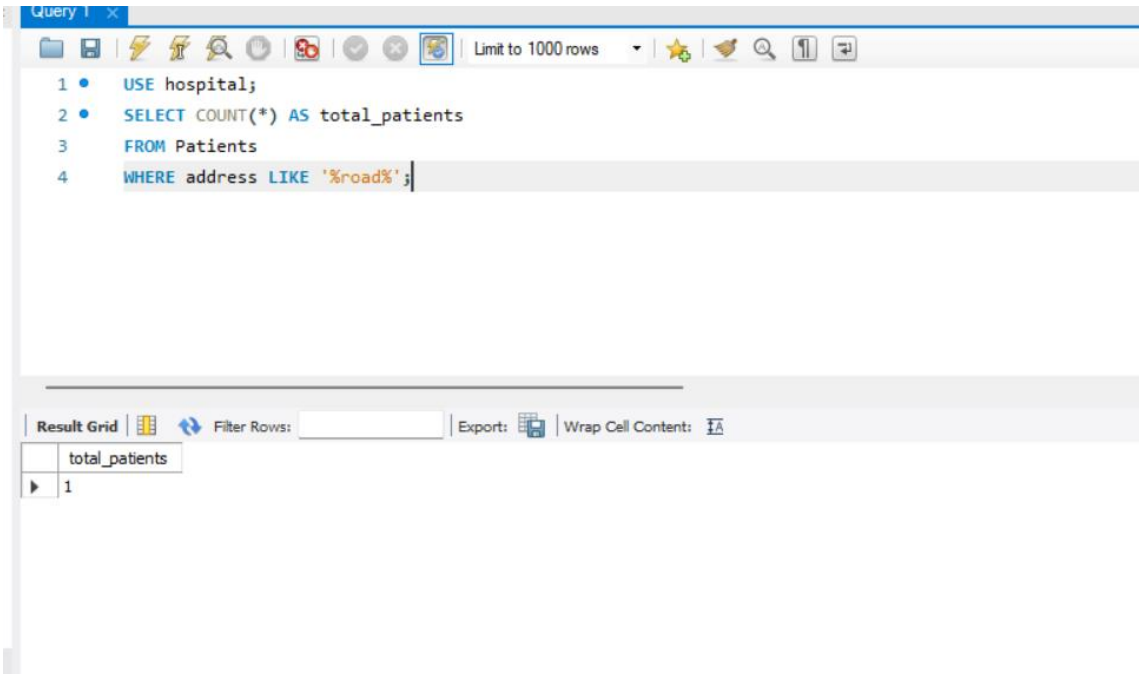
15. Set the chief executive officer (CEO) of the hospital to be called Michael Dean. (2 Marks)



16. Show all staff members who have been working there for longer than 3 years. (2 Marks)

17. Show the detail of all patient names, medical history, appointment dates, times, and treatment (Hint: join operation). (2 Marks)

18. Identify how many patients have the word road in their address. (1 Marks)



In this exercise I used COUNT in order to

19. Create a view that will show the result of a query drawing information from three tables at once. (3 marks)

20. Drop all information contained in the appointments relation. (2 Marks)

